Piecewise-Affine Local Volatility

Calibrating a Dupire surface from sparse quotes: model, PDE, gradients, and speed

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Abstract

This note documents the Vol-Fitter's full local-volatility model: a continuous piecewise-affine (P_1 finite-element) local-variance surface $\nu_\theta(t, x)$, calibrated directly to sparse vanilla and variance-swap quotes through the forward Dupire PDE as an implicit pricing map. Unlike the slice models (Notes 01–03) this is a genuine two-dimensional local-volatility surface: strict positivity of the nodal variances makes the implied prices butterfly- and calendar-clean up to scheme noise, which the model *measures* rather than assumes. We derive the normalized forward Dupire equation, the P_1 parametrization and its positivity guarantee, the implicit-Euler discretization and observation operator, the variance-swap log-contract replication and its faster source-PDE alternative, and the calibration objective with its roughness, convex-wing and front-tie regularizers. We then give the sensitivity PDE and the adjoint gradient, and — the engineering heart of the note — the numerical optimizations that took the cold fit from $\sim 86\,\mathrm{s}$ to single-digit seconds: a compiled vectorized-Thomas Dupire march ($\sim 6.5\times$), a stall-based early-stop, a parametric Dupire warm seed, and the matrix-free Gauss–Newton solver that avoids the dense SVD. A synthetic round trip recovers a known surface to 0.6 vol bps, and the production fit on a static Bloomberg fixture reaches 2.8 vol bps on SPY and 11.7 on NVDA. The note closes with the short-dated-fit fixes and the convex-wing regression that the benchmark caught.

Contents

1 The idea

A slice model fits one expiry at a time and stitches expiries together with a calendar constraint. A *local-volatility* model does the opposite: it posits a single function $\sigma_{\text{loc}}(t, S)$ such that, under $\mathrm{d}S_t = \sigma_{\text{loc}}(t, S_t)\, S_t\, \mathrm{d}W_t$ (in forward-normalized units), the model reproduces *all* expiries and strikes at once. Dupire showed such a function exists and is unique given an arbitrage-free price surface. The Vol-Fitter follows the Andreasen–Huge philosophy: rather than *differentiate* a fitted price surface to extract σ_{loc} (numerically explosive in sparse data), it *parametrizes* σ_{loc} directly and prices *forward* through the Dupire PDE, calibrating the parameters to quotes.

Heuristic

Extracting local vol by the Dupire formula $\sigma^2_{\text{loc}} = \partial_T C / (\frac{1}{2} K^2 \partial_{KK} C)$ divides two noisy numerical derivatives of a sparse surface — a recipe for garbage in the wings. Calibrating local vol *forward* instead never differentiates the data: it asks "what positive surface, run through the PDE, best reprices the quotes?" Positivity of the surface is imposed on the *parameters*, so every iterate is arbitrage-clean by construction — the same philosophy as LQD, one dimension up.

2 The forward Dupire equation

Work in forward-normalized strike $x = K/F_T$ and let $c(T, x)$ be the normalized call. Writing local variance $\nu(t, x) = \sigma^2_{\text{loc}}$, the forward Dupire equation is

$$\boxed{\partial_T c(T, x) = \tfrac{1}{2}\, \nu(T, x)\, x^2\, \partial_{xx} c(T, x), \qquad c(0, x) = (1 - x)^+.} \tag{1}$$

It is a forward parabolic PDE in (T, x) with the call payoff as initial data. Two facts make it the right object. First, $\partial_{xx} c \geq 0$ is exactly the butterfly (density) condition, and $\nu \geq 0$ keeps the equation parabolic, so a positive ν *propagates* non-negativity of the density forward in maturity. Second, monotonicity of c in T (the calendar condition) follows from (1) whenever $\nu \geq 0$, since the right-hand side has the sign of $\partial_{xx} c \geq 0$. Thus a strictly positive local-variance surface is *structurally* free of both static arbitrages, up to the discretization noise the model measures (section 4.2).

3 The piecewise-affine surface

3.1 Parametrization

The local variance is a continuous piecewise-affine (P_1) function on a tensor grid of $n_t \times n_x$ vertices (τ_i, ξ_j),

$$\nu_\theta(t, x) = \sum_{i,j} \theta_{ij}\, \phi_{ij}(t, x), \tag{2}$$

where ϕ_{ij} are the bilinear (or barycentric/Delaunay) hat functions and $\theta_{ij} = \nu(\tau_i, \xi_j) \geq 0$ are the *nodal local variances* — the calibration parameters. The flat vector is $\theta = \mathrm{vec}(\theta)$, time-major.

Proposition 1 (Nodal positivity ⇒ surface positivity). *A P_1 function is a convex combination of its vertex values on each cell. Hence if $\theta_{ij} \geq 0$ for all i, j, then $\nu_\theta(t, x) \geq 0$ everywhere, and $\nu_\theta > 0$*

if the nodal values are bounded below by a positive constant.

This is why the calibration enforces only *box bounds* on θ (default local-variance bounds $[0.005, 0.20]$, i.e. local vol in $[7\%, 45\%]$): positivity of the whole surface reduces to positivity of finitely many numbers. Beyond the lowest strike vertex the surface extrapolates the local variance *linearly* with a slope set by the first interior cell (`left_extrap_a`; Note 09), rising toward $x \to 0$; the right wing is flat-clamped.

3.2 Grid design

The vertex grid is delta-spaced in strike and floored per expiry so that even a narrow short-dated smile lands enough vertices on its curvature (`gridXNodes`, `gridXMinPerExpiry`; section 9). The time vertices (`gridTNodes`) carry the term structure. The grid is deliberately modest — a dozen strike vertices, ten in time — because each vertex is a free parameter and the PDE solve cost scales with the grid; the regularizer (section 6) supplies the smoothness that a coarse grid would otherwise lack.

4 The pricing map

4.1 Discretization

The PDE (1) is solved on a uniform spatial grid x_g (default ~ 1201 nodes spanning several standard deviations, `DEFAULT_N_K`) by fully implicit (backward) Euler in maturity with step $\leq$ `DEFAULT_DT_MAX` $= 1/400$, optionally Rannacher-started (a few implicit quarter-steps before Crank–Nicolson). Each step solves a tridiagonal M-matrix system

$$(I - \Delta T\, A_n)\, U^{n+1} = U^n, \tag{3}$$

where A_n encodes $\frac{1}{2}\nu\, x^2 \partial_{xx}$ at step n. The matrix is an M-matrix (diagonally dominant, non-positive off-diagonals), so the scheme is unconditionally stable and monotone — it cannot create spurious negative densities. The observation operator reads the normalized call at each quote's (T, x) off the solution by interpolation.

4.2 No-arbitrage diagnostics

Because the surface is only *up to scheme noise* arbitrage-free, the model measures the residual: a butterfly tolerance on $\partial_{xx} c$ and a calendar tolerance on $\partial_T c$ (both 10^{-8}). The model is gated behind these diagnostics rather than assuming cleanliness — a deliberate design choice.

5 Variance swaps

A variance-swap quote is the fair strike of realized variance, which for a diffusion equals the expected total variance to T. Two pricers are available. The *static* method uses the log-contract replication

$$I(T) = 2\int_0^1 \frac{P(T, x)}{x^2}\, \mathrm{d}x + 2\int_1^\infty \frac{C(T, x)}{x^2}\, \mathrm{d}x, \tag{4}$$

a trapezoid over the priced surface (half-width and node count `VS_HALF_WIDTH`, `VS_POINTS`). The faster *source-PDE* method (default-off, gated by `varSwapMethod`) solves one backward source PDE $\partial_t g + \frac{1}{2}\nu x^2 \partial_{xx} g + \nu = 0$, $g(T,\cdot) = 0$, and reads $I(T) = g(0,1)$ — with analytic sensitivities $\partial I/\partial\theta$ from the same multi-RHS machinery, and without the $x_{\max}$-dependence of the replication tail. Note 08 develops both in full.

6 The calibration objective

The fit minimizes a stacked least-squares residual over θ,

$$\min_{\theta\in[\nu_{\mathrm{lo}},\nu_{\mathrm{hi}}]} \tfrac{1}{2}\Big(\underbrace{\|r_{\mathrm{opt}}\|^2}_{\text{quotes}} + \underbrace{\|r_{\mathrm{vs}}\|^2}_{\text{var-swaps}} + \underbrace{\lambda\|L(\theta-\theta_{\mathrm{ref}})\|^2}_{\text{roughness}} + \underbrace{w_{\mathrm{cvx}}\|r_{\mathrm{cvx}}\|^2}_{\text{convex wing}} + \underbrace{w_{\mathrm{ft}}\|r_{\mathrm{ft}}\|^2}_{\text{front tie}}\Big), \tag{5}$$

with the blocks:

- **Option residuals** $r_{\mathrm{opt},q} = (P_q(\theta) - y_q)/\eta_q$ in vega-normalized forward-call price units (mid), or the bid–ask/haircut band hinge with a small mid anchor when band edges are present (Note 07).
- **Var-swap residuals** $r_{\mathrm{vs}} = (I(\theta) - z)/\zeta$ (Note 08).
- **Roughness** $\sqrt{\lambda}\,L(\theta - \theta_{\mathrm{ref}})$ with L the spacing-aware second-difference operator and θ_{ref} a reference surface; the time-vs-strike balance is `gridRegRho`. This is the smoothness that lets a coarse grid generalize.
- **Convex wing** hinge: a one-sided penalty enforcing local convexity of the wing, confined to vertices *below the deepest quote* (the extrapolation tail only — see section 9.1).
- **Front tie**: a light penalty tying the front-end vertices to the ATM level where the shortest expiry has little strike coverage.

7 Sensitivities and the adjoint

7.1 The sensitivity PDE

Each parameter θ_ℓ enters (1) linearly through ν_θ, so the price sensitivity $s_\ell = \partial c/\partial\theta_\ell$ obeys the same parabolic PDE with a *source*:

$$\partial_T s_\ell = \tfrac{1}{2}\nu_\theta x^2 \partial_{xx} s_\ell + \tfrac{1}{2}\phi_\ell\, x^2 \partial_{xx} c, \qquad s_\ell(0,\cdot) = 0. \tag{6}$$

Discretized, every s_ℓ shares the *same* tridiagonal factor as the value solve (3), differing only in the source — so the Jacobian columns are a single multi-RHS back-substitution against one factorization. This is the forward-sensitivity route, $O(N_t N_x m)$ for m active parameters.

7.2 The adjoint gradient

When only the *gradient* of the scalar objective is needed (not the full Jacobian), the discrete adjoint computes it in one forward value solve plus one backward adjoint sweep, $O(N_t N_x)$ *independent of* m. The Vol-Fitter uses the forward sensitivities for the Gauss–Newton Jacobian (it needs the columns) and keeps the adjoint as the route for very large vertex counts.

8 Solvers and the speed story

The cold fit of (5) was, at ~ 533 vertices, an ~ 86 second affair hitting the 200-evaluation cap — the single heaviest computation in the app. Driving it down is most of the engineering in this model, and the lessons are worth stating because several plausible ideas *failed*.

8.1 Where the time goes

On a cold fit the cost splits roughly as: optimizer linear algebra / dense SVD $\sim 52\%$, PDE sensitivity march $\sim 32\%$, Jacobian assembly $\sim 14\%$, value solve $\sim 2\%$. No single per-evaluation lever moves the total, because the cost is distributed — which is why four separate "faster per evaluation" attempts (coarse grid, first Numba try, Rannacher, and the first matrix-free GN) each won only ~ 1.1–$1.2\times$ and were shelved.

8.2 What worked

Performance

The cold fit is now ~ 3–$6\times$ faster than the original banded baseline (scaling with grid size); recalibrations are near-instant via warm start. The levers, in order of impact:

1. **Stall-based early-stop.** Track the option-block misfit and stop at the best iterate once it stalls. Because fewer evaluations multiply march, assembly *and* optimizer together, this scales the *whole* fit: measured $1.45\times$ (SPY) to $3.3\times$ (NVDA), at +0.1–0.25 vol bp.

2. **Compiled vectorized-Thomas march.** A `@njit` no-pivot Thomas solver with the sensitivity columns as the contiguous SIMD inner loop and a fused source: $\sim 6.5\times$ the LAPACK banded march, numerically exact to $\sim 10^{-15}$. The first Numba attempt (column-outer scalar Thomas) won only $1.2\times$ — the real lever was the loop order.

3. **Matrix-free Gauss–Newton.** The default solver avoids `trf`'s dense SVD (52% of an evaluation) by solving the GN step with preconditioned LSMR on matrix-free $Jv/J^{\top}w$ products. Once the march is cheap, the no-SVD evaluations win: ~ 1.3–$1.65\times$ over TRF. Gated to the smooth mid-fit target.

4. **Parametric Dupire warm seed.** Seed θ from the parametric surface's local variance (~ 1.3–$1.8\times$ on cold fits) and from the previous surface on recalibration ($\sim 38\times$).

The order-of-magnitude wins are spent; remaining levers are incremental.

Caution

Removing the dense SVD made fits *slower* at ≤ 440 vertices, because there the bottleneck is the PDE march, not the SVD — the SVD wall is a $\gtrsim 1000$-vertex (future non-tensor "bowtie" grid) phenomenon. The clean synthetic benchmark (zero residual, in bounds) hid this: GN converged in 8 evaluations there but did *not* converge within the cap on stiff real data. The honest lesson: measure on real chains, and distrust a synthetic-only speed-up.

9 Short-dated calibration

A genuine one-week smile fit *catastrophically* (108 vol bp RMS vs the parametric ~ 47) while normal expiries fit well. A measure-first diagnosis (per-expiry side-channel counting in-range vertices, vega flooring and PDE steps) ruled out the usual suspects — vega floor, time steps, local-vol cap, prior leakage — and isolated *strike under-resolution*: the delta strike axis is sized to the *longest* expiry and clipped to the global range, so a narrow short smile landed only ~ 3 vertices on its sharpest curvature. Two fixes:

1. **Per-expiry coverage floor** (`gridXMinPerExpiry`, default 8): after the delta axis is built, split the widest in-range gaps until each expiry has at least $m_{\min}$ vertices inside *its own* traded range — adding nodes only to under-covered short fronts.

2. **Short-expiry-aware PDE step**: refine the shared PDE x-step to $0.3\times$ the smallest ATM $\sigma\sqrt{\tau}$, snapped so $x = 1$ stays a node, clamped to [1/400, 0.01]. Normal surfaces stay on 0.01 (byte-identical).

The one-week fit fell from $108 \to 23.5$ vol bp (now *better* than the parametric), with the long expiries byte-identical.

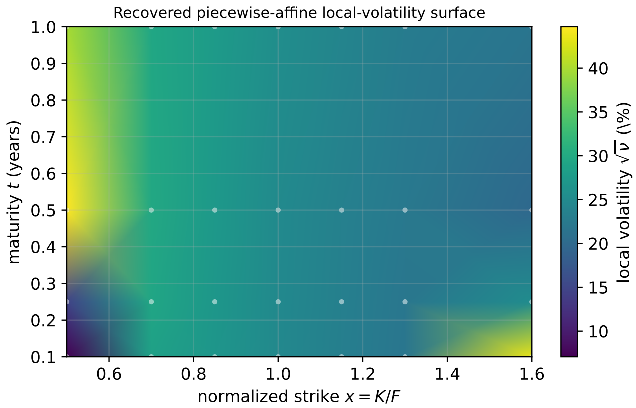
9.1 The convex-wing regression

The benchmark also caught a subtler bug. The convex-wing constraint originally selected *every* vertex in the deep wing regardless of data; at a user's saved `gridXNodes` = 20 it stacked convexity penalties onto densely-quoted put strikes and forced the *wrong* wing on low-vol SPY (NVDA's naturally convex wing had hidden it). The fix confines `convex_cols` to vertices *below the deepest quote* — the extrapolation tail only. SPY went $25.7 \to 2.6$ vol bp. The lesson echoes Note 09: a shape constraint must not be imposed where the data already speaks.

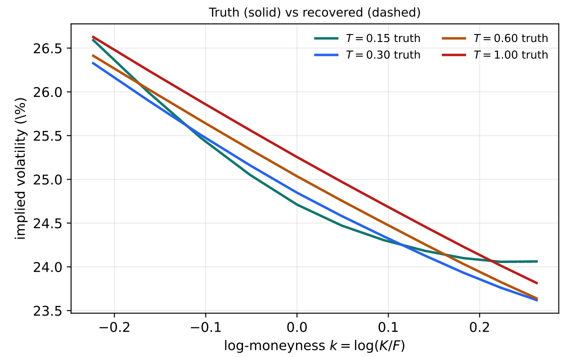
10 Worked examples

10.1 Synthetic round trip

A known skewed local-variance surface is priced through the forward PDE, quotes are generated at four expiries, and the surface is recovered from a *flat* seed by the production calibrator. Figure 1 shows the recovered surface and the implied-vol fit: the calibrator recovers the truth to a maximum of 0.6 vol bps in 33 evaluations (0.92 s wall, 28 vertices), with the skew (higher local vol at low strike) faithfully reconstructed.



(a) recovered local-vol surface



(b) IV fit, truth vs recovered

Figure 1: Synthetic round trip through the real calibrator: the skewed surface is recovered to 0.6 vol bps from a flat seed.

10.2 The Bloomberg benchmark

On the static Bloomberg fixture the production fit reaches the quality in table 1 and fig. 2: 2.8 vol bps surface RMS on SPY and 11.7 on NVDA, with no node at a bound on the easy expiries. The harder NVDA short-dated expiries carry the larger error — the regime section 9 addresses.

Table 1: Production local-vol fit on the Bloomberg fixture (fresh run).

Name	surface RMS (bp)	worst expiry (bp)
SPY	2.8	4.4
NVDA	11.7	12.4

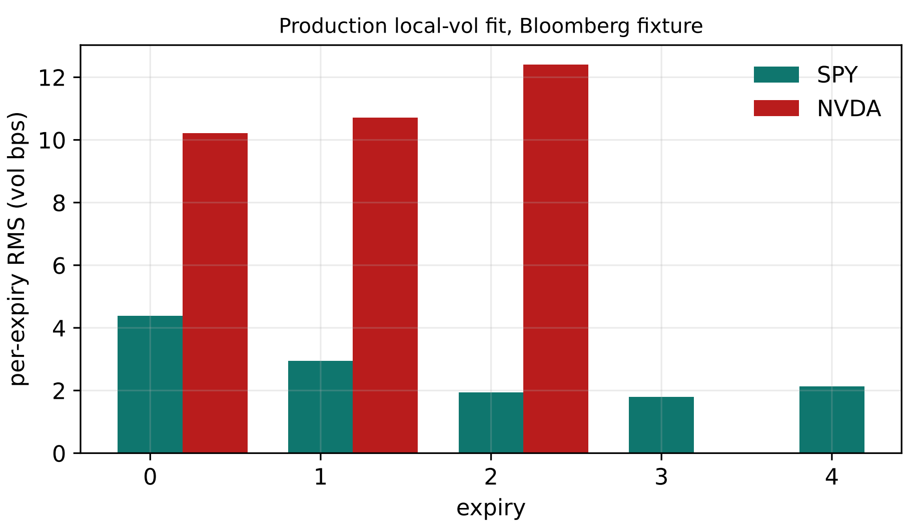


Figure 2: Per-expiry weighted RMS for SPY and NVDA on the fixture.

11 What is original, and limitations

The P_1-positivity guarantee and the forward-calibration philosophy are Andreasen–Huge; the contributions here are engineering depth. The *vectorized-Thomas Numba march* (loop order as the lever, 6.5×), the *stall early-stop* that scales the whole fit, the *matrix-free GN* correctly scoped to where the SVD actually dominates, the *source-PDE var-swap* with analytic sensitivities, and the *measure-first* short-dated diagnosis are each non-obvious and were validated against a real benchmark that twice caught regressions a synthetic would have missed. Limitations: the model is a pure diffusion (no jumps), so a true short-dated event spike is only approximated; the tensor grid wastes vertices in the corners (the future non-tensor bowtie grid is where GN's no-SVD advantage would finally dominate); and the fit, though much faster, remains the heaviest single computation in the app.

A Hyperparameter atlas

Table 2: Local-volatility hyperparameters.

Knob	Default	Role
Surfaced (OptionsSettings)		
`gridXNodes`	12	Strike vertices of the variance surface.
`gridXMinPerExpiry`	8	Min in-range strike vertices per expiry (section 9).
`gridTNodes`	10	Maturity vertices.
`gridRegLambda` λ	10^{-2}	Roughness penalty strength.
`gridRegRho` ρ	1.0	Time-vs-strike roughness balance.
`convexWingWeight`	10^{3}	Convex-wing hinge weight (tail only).
`frontTieWeight`	10^{-2}	Front-end tie-to-ATM weight.
`lvVolCapMult`	3.0	Local-vol cap as a multiple of max implied vol.
`leftWingSlopeMult`	1.5	Left local-variance extrapolation slope multiple.
`varSwapMethod`	`static`	Var-swap pricer: replication or source-PDE.
`lvSolver`	`gn`	Gauss–Newton (default) or `trf`.
`lvFastKernel`	`true`	Numba vectorized-Thomas march.
`lvEarlyStop`	`true`	Stall-based early stop.
`timeScheme`	`implicit`	Implicit Euler or Rannacher/CN.
Hidden (PDE / solver)		
`DEFAULT_N_K`	1201	PDE spatial nodes.
`DEFAULT_DT_MAX`	1/400	PDE max time step.
`N_RANNACHER`	4	Rannacher start quarter-steps.
`SPAN_SD` / `SPAN_MIN`	7.5 / 0.6	PDE grid span (sd / floor).
`VAR_FLOOR`	10^{-6}	Dupire local-variance floor.
variance bounds	$[0.005, 0.20]$	Nodal local-variance box (vol $[7\%, 45\%]$).
`x_scale`	`jac`	Optimizer column scaling.
`xtol/ftol/gtol`	10^{-8}	Optimizer tolerances.
`max_nfev`	200	Evaluation cap.

Knob	Default	Role
`stall_window/rtol`	12–18 / ~ 3–5×10^{-3}	Early-stop window/tolerance.

B Performance notes

1. **Numba march** (`lvFastKernel`). No-pivot factor-once Thomas with the m sensitivity columns as the contiguous inner SIMD loop and fused source: 6.1–6.9× vs LAPACK `dgbsv`, exact to $\sim 10^{-15}$. Graceful banded fallback if Numba is absent.

2. **Early stop** (`lvEarlyStop`). Best-iterate stop on option-block stall; scales the whole fit, 1.45× (SPY) to 3.3× (NVDA), +0.1–0.25 vol bp. `stall_window`= 0 is byte-identical.

3. **Matrix-free GN** (`lvSolver=gn`). Preconditioned LSMR on $Jv/J^{\mathsf{T}}w$; avoids the dense SVD. 1.3–1.65× over TRF at tensor-grid sizes; gated to the smooth mid target.

4. **Warm starts.** Previous-surface seed ($\sim 38\times$ recal) and parametric Dupire cold seed (1.3–1.8×).

5. **Shelved (documented):** coarse grid (biases θ), first Numba attempt (wrong loop order), Rannacher (1.1× + arb risk), GN for the non-smooth band objective, thread/process parallelism (GIL). Four distributed-cost dead-ends confirming the cost is spread, not localized.

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